

BAYESIAN STOCHASTIC SEARCH VARIABLE SELECTION (BSSVS) IN PHYLOGEOGRAPHIC MODELS

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PhylogeogRaphy Workshop

Day 4

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A QUICK RECAP SO FAR:

We've studied phylogeography through discrete trait analysis

- 1 for a fixed tree, we can do joint or marginal reconstructions.
 - I ace (joint and/or marginal reconstructions)
 - II simmap (joint and/or marginal reconstructions + character mapping)
 - III SSE models (BiSSE, MuSSE, SAASI, etc)
- 2 while jointly estimating a phylogeny from sequence data – gives us a collection of character-mapped trees

We also learned how to carry out a full phylodynamics analysis

- 1 multi-type birth death model – gives us a collection of character-mapped trees
- 2 structured coalescent model – also gives a collection of character-mapped trees

THIS MORNING:

We take a step back from phylodynamics to return to Bayesian discrete trait analysis,

$$p(\theta, \mathbf{Q}, \mathbf{A} | \mathbf{s}, \mathbf{x}) \\ \propto \int_{\mathcal{T}} \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}, \mathbf{A}) \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} | \mathcal{T}, \mathbf{Q}) p(\mathcal{T} | \theta) p(\theta) p(\mathbf{Q}) p(\mathbf{A})$$

THIS MORNING:

The problem: in many situations, we may collect data from N locations, but only a small number of transitions between locations occur. The stochastic rate matrix for geographic movement,

$$\mathbf{Q} = \begin{pmatrix} -\sum_i q_{i1} & q_{12} & q_{13} & \dots & q_{1N} \\ q_{21} & -\sum_i q_{i2} & q_{23} & \dots & q_{2N} \\ q_{31} & q_{12} & -\sum_i q_{i3} & \dots & q_{3N} \\ \dots & \dots & \dots & \dots & \dots \\ q_{N1} & q_{N2} & q_{N3} & \dots & -\sum_i q_{iN} \end{pmatrix}$$

will have many entries $q_{ij} = 0$. Our estimates $\hat{\mathbf{Q}}$ are expected to have high variance because there are few migration events but many parameters to estimate¹.

¹Philippe Lemey et al. "Bayesian phylogeography finds its roots". In: *PLoS computational biology* 5.9 (2009), e1000520.

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CONNECTION TO REGRESSION

A similar problem can arise in linear regression. We will port over the solution from that context.

In linear regression, a statistical model has the form

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon,$$

where $\mathbf{Y}$ is an n -dimensional vector, $\mathbf{X}$ is an $n \times p$ matrix of covariates, β is a vector of p regression coefficients, and ϵ is an n -dimensional vector of mean-zero Gaussian errors.

CONNECTION TO REGRESSION

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

If the number of possible predictors in β is large but only a small number of them are actually nonzero, the problem boils down to identifying the small set of predictors β_i that should be included in the model (and setting the rest to zero)

REGRESSION

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

Regularization techniques like LASSO or ridge regression are designed to impose penalties on the estimation procedure to ensure that spurious predictors β_i are not included in the model.

LASSO

A linear regression model,

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon,$$

would typically be fit through least squares,

$$\min_{\beta} \sum_i \left(y_i - \beta_0 - \sum_j x_{ij} \beta_j \right)^2,$$

but a regularization technique would fit according to a different criterion, based on a regularization parameter λ , by minimizing

$$\min_{\beta} \sum_i \left(y_i - \beta_0 - \sum_j x_{ij} \beta_j \right)^2 + \lambda \sum_j |\beta_j|.$$

As λ increases, more of the coefficients $\beta_j \rightarrow 0$.

LASSO

That is, the fitting criterion

$$\min_{\beta} \sum_i \left(y_i - \beta_0 - \sum_j x_{ij} \beta_j \right)^2 + \lambda \sum_j |\beta_j|,$$

balances model fit (sum of squares) while keeping the number of nonzero coefficients β_j small (or “parsimonious”).

This identifies the most important predictors β_j .

LASSO vs. RIDGE REGRESSION

LASSO fits models by incorporating the L_1 norm in the objective function:

$$\min_{\beta} \sum_i \left(y_i - \beta_0 - \sum_j x_{ij} \beta_j \right)^2 + \lambda \sum_j |\beta_j|.$$

Another approach is “ridge regression”, which incorpoates the L_2 norm in the objective function:

$$\min_{\beta} \sum_i \left(y_i - \beta_0 - \sum_j x_{ij} \beta_j \right)^2 + \lambda \sum_j \beta_j^2.$$

LASSO vs. RIDGE REGRESSION

Minimizing $\sum_j |\beta_j|$ (LASSO) performs “feature selection” because a subset of the parameters $\beta_j = 0$ in the fitted model.

Ridge regression minimizes $\sum_j \beta_j^2$. In the fitted model, several of the $\beta_j \ll 1$, but do not actually equal zero. Ridge regression penalizes over-fitting but still retains all of the predictors.

Recall: in Euclidean space, points that are a fixed distance from the origin under the L_2 norm lie on a circle/sphere centered on the origin. Under the L_1 norm, points that are a fixed distance from the origin lie on diamonds/cubes, the vertices of which fall on the axes.

BACK TO OUR PHYLOGEOGRAPHY PROBLEM

In the phylogeography context, estimating a high-dimensional but sparse stochastic rate matrix, $\mathbf{Q}$, is a similar type of problem

We want to penalize our likelihood for including too many parameters q_{ij} that are not well-supported by the data. This is the problem that BSSVS is designed to solve (actually BSSVS was originally developed for regression, but Lemey et. al. adapted it for the phylogeographic context)²

We consider symmetric migration rates, $q_{ij} = q_{ji}$

²Philippe Lemey et al. "Bayesian phylogeography finds its roots". In: *PLoS computational biology* 5.9 (2009), e1000520.

BSSVS³

The BSSVS strategy we will discuss is similar to LASSO: first, augment the parameter space with a set of binary indicator variables, δ_{ij} for each of the $p = N(N - 1)/2$ migration rate parameters, q_{ij} , where $\delta_{ij} = 0$ or 1.

Then, impose a δ -dependent prior distribution on the migration rate parameters, $p(q_{ij}|\delta_{ij})$, that is

- I centered at 0
- II has zero variance when the corresponding $\delta_{ij} = 0$

³Philippe Lemey et al. "Bayesian phylogeography finds its roots". In: *PLoS computational biology* 5.9 (2009), e1000520.

INCORPORATE INTO MCMC

Before, we performed inference on the posterior distribution

$$p(\theta, \mathbf{Q}, \mathbf{A} | \mathbf{s}, \mathbf{x}) \\ \propto \int_{\mathcal{T}} \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}, \mathbf{A}) \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} | \mathcal{T}, \mathbf{Q}) p(\mathcal{T} | \theta) p(\theta) p(\mathbf{A}) p(\mathbf{Q})$$

INCORPORATE INTO MCMC

Now, we augment our posterior with $p(\delta)$ and proceed as we normally would in MCMC:

$$p(\theta, \mathbf{Q}, \mathbf{A} | \mathbf{s}, \mathbf{x}) \\ \propto \int_{\mathcal{T}} \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}, \mathbf{A}) \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} | \mathcal{T}, \mathbf{Q}) p(\mathcal{T} | \theta) p(\theta) p(\mathbf{A}) p(\mathbf{Q} | \delta) p(\delta)$$

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ADDING COVARIATES

The strategy of augmenting the model with binary indicators δ and estimating them through MCMC suggests other possibilities:

- the q_{ij} could depend on other covariates
- conceptually similar to modeling time-dependent rates with piecewise constant functions (think: Bayesian skyline models in BEAST2)